UGH

Are you living in a filter bubble?
The flowchart shown on this website will tell you if you are actually living in a filter bubble. Try it out. **https://dgit.in/mar20-43**

WOW

Pixel-binning explained
Understand what pixel-binning is and how it makes a difference in photography on smartphones. **https://dgit.in/mar20-44**

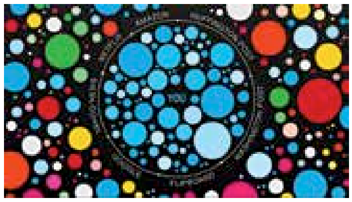
Filter Bubbles

The nature of the algorithms that deliver you advertisements

Aakanksha Chaturvedi |
feedback@digit.in

Ever wondered why one minute you are searching for a recipe and the next moment you are bombarded with advertisements of food delivery apps ? This happens because website algorithm selectively filters information using over 57 parameters like search history, cookie data, click behaviour, personal information ,etc to personally tailor your search results.

At first glance, personalised search results might seem harmless, rather convenient but a closer look shows that they lead to a state of intellectual isolation termed as 'Filter Bubble' as coined by internet activist Eli Pariser. Filter Bubbles effectively isolate the user into their own cultural and ideological bubbles, showing results and news items which conform only to their own personal opinion on a subject, missing out the big picture. Filter Bubbles also lead to the formation of 'Echo Chambers' wherein similar ideas are repeated reinforcing them in the minds of the readers and eliminating any opposite views.



Before the advent of the internet, this job of 'filtering out' or 'gate keeping' was done by journalists and editors. The arrival of internet meant that the information of the whole world was at the tip of your finger. But the algorithms, unlike journalists, lack ethics, morals and conscience. Unaware of the constant censoring search engines do, people accept what's shown to them as the truth.

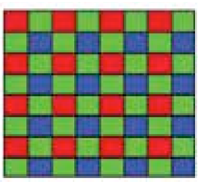
This invisible curation traps people in their own bubble of auto-propoganda, indoctrinating them with their own ideas and leaving no space for disagreement.

Though it seems inescapable, individuals can counter this problem by using search engines like Qwant, Disconnect, Searx, DuckDuckGo, etc. Users can also consciously seek out contrasting views to try to get the bigger picture about an issue.

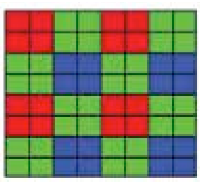
Does megapixel matter?

On the Digit Forum, *@SoorajDigit* asked if it make sense to have 48MP, 64MP, 108MP main cameras with 3 or 4 extra cameras? *@Anorion* answers. MegaPixel does not really matter. If you go just by that then LeEco Le Max 2 or something had the biggest MP count in its range, and top of the line specs, but was still a trash phone. Most smartphones use sensors by either Samsung or Sony. Apart from the top of the line sensors, the specifications are similar and even interchangeable, so some entry level phone models may have some sensors from Sony, and others by Samsung.

Now, how can these manufacturers offer increasing MP size in smaller sensors? It is because each individual imaging element (pixel) on the sensor is getting smaller. This does not translate to better picture quality. Think of the shower of photons that leads to getting an image as a rainfall. You can have a large array of big buckets (dSLRs), a small array of big buckets (bridge/prosumer), or a small array of small cups (smartphones). The imaging quality is dependent on the total amount of light or water captured. Although the smartphones have more cups (higher MP), the bigger dSLRs capture more light (or rain) with a reduced MP count. HTC UltraPixel sensors have individual imaging elements that are more than twice the size of competition, and are able to capture more light, and deliver a better image quality.



Bayer Filter



Quad-bayer Filter

The next important thing to understand is the arrangement of the individual pixels on the sensors. Older phones used a quad bayer filter, which was an array with one red pixel, one blue pixel, and two green pixels. On newer devices, smaller sensors meant that smartphone companies had to cheat and came up a quad bayer array, where the pixels are arranged in a matrix of four imaging elements replacing what was previously a single pixel. This means that the block of four pixels can behave like a single pixel when required, allowing for better low light performance, and dynamic range. Both Samsung and Sony use the same arrangement. Does it improve performance? Yes, in some cases, but only because they need to do that due to tiny sensors, larger batteries, multiple lenses, selfie shooters and notched displays.

So, what are the main points? 1) The amount of light falling on the sensor decides the image quality, not the MP count. 2) Notched displays, multiple lenses, etc are exerting pressure on smaller sensors with increased MP count. While quality of images are improving, the progress is slow due to these presssures. 3) The best cameras are in the top-end Samsung, Huawei, Apple and Google devices.

THE DIGIT COMMUNITY IS FILLED WITH BRIGHT IDEAS and we find the best among the best to bring them to you in this section.